

# Prajwal Sah

+977-9808899736 | prajwalsah726057@gmail.com | linkedin.com/in/prajwal-sah | github.com/Iam-Prajwal |

**Builder by instinct | Engineer by discipline | Poet by passion**

I build across the boundaries of software engineering, AI/ML, and cybersecurity – and lead technical communities as a student ambassador.

## Education

**Khwopa Engineering College, Bhaktapur**

*Bachelor of Engineering in Computer Engineering*

2023 – Present

*Expected Graduation: Mar 2027 | CGPA: 3.70/4.0*

**Khwopa Higher Secondary School, Bhaktapur**

*Higher Secondary Education (+2 Science)*

2022

*GPA: 3.50*

**Sainik Awashiya Mahavidyalaya, Bhaktapur**

*Secondary Education Examination (SEE)*

2020

*GPA: 3.80*

## Experience

**Vortex Tech**

*Cybersecurity Intern (Project-Based)*

Jul 2026 – Aug 2026

*Remote*

- Identified open ports, active services, and potential attack surfaces across test environments, documenting results in a structured security assessment report, by conducting Nmap-based network reconnaissance and vulnerability scanning.
- Improved end-user password security awareness, translating findings into actionable recommendations, by developing a password strength checker that evaluated key password characteristics.
- Built foundational offensive-security competency, validated through weekly technical assignments and documentation, by applying Linux, networking, and reconnaissance best practices in hands-on exercises.

**Bhaktapur Municipality (Collaboration through College)**

*Software Developer*

Jan 2026 – May 2026

*Live Website*

- Delivered a full-stack tourism and digital heritage platform officially adopted and publicly hosted by Bhaktapur Municipality, by building **15+ responsive pages** and **15+ service modules** with Django, HTML, CSS, JavaScript, Bootstrap, and PostgreSQL.
- Enabled location-aware discovery of nearby attractions with configurable search radii, powering interactive maps for platform users, by implementing secure authentication and **REST API** integrations for geolocation features.
- Shipped the platform across the full development lifecycle within a **4-member team**, meeting municipal deployment requirements, by leading requirement analysis, implementation, testing, and stakeholder feedback cycles.
- Achieved production readiness ahead of public launch, resulting in official municipal adoption, by owning backend integration, database design, and deployment.

## Research and Ongoing Works

- Developing a 1-bit encoder for Devanagari characters, exploring compact binary representations for efficient character encoding and processing.
- Researching binary encoding techniques for Devanagari text, focusing on representation efficiency, character coverage, and computational simplicity.
- Evaluating the proposed encoding approach across Devanagari character sets to assess encoding consistency, storage efficiency, and practical applicability.

## Technical Skills

**Languages:** Python, C, C++, JavaScript, HTML, CSS, ReactJS (Workshop/Practical Exposure)

**Data Structures & Algorithms:** Arrays, Linked Lists, Trees, Graphs, Dynamic Programming

**Frameworks & Libraries:** Django, Flask, REST APIs, Bootstrap, Tailwind CSS

**AI & Machine Learning:** LLM APIs, Retrieval-Augmented Generation (RAG), Agentic AI, NLP, Prompt Engineering

**Databases:** PostgreSQL, MySQL, MongoDB

**Cybersecurity Tools:** Kali Linux, Wazuh SIEM, MITRE ATT&CK, Wireshark, Nmap

**Tools & Fundamentals:** Git, GitHub, LaTeX, WordPress, Agile/Scrum, Testing & Debugging, Problem Solving

## Projects

**Log Semantic Augmentation - Agent**

*Final year Project*

- Developing LSA-Agent, a privacy-preserving Agentic RAG framework, to automate security incident investigation by combining Wazuh SIEM data with local LLMs.
- Researching AI-assisted threat hunting and on-premise cybersecurity analytics by mapping detection workflows to the MITRE ATT&CK framework in privacy-preserving environments.
- Evaluating local large language models for cybersecurity incident analysis and alert triage to assess their effectiveness against cloud-based alternatives.

## Fallacy Detector & AI Reasoning Assistant

*Hackathon 2025 – AI/ML Track Winner*

*Github*

- Achieved **82% classification accuracy** across **13 fallacy categories** on **1,000+ annotated arguments**, by fine-tuning an ELECTRA transformer model integrated with OpenRouter LLM APIs.
- Delivered real-time reasoning assistance directly in the browser, supporting analysis of both selected webpage text and user-authored content, by building a **Manifest V3** browser extension.
- Won the **AI/ML Track** at KU Hackathon 2025 against university-wide competition, by architecting, developing, and presenting the solution within a **4-member team** over a **48-hour** hackathon.

## Spotify Clone

*Personal Project*

*Github*

- Developed a full-stack music streaming web application using Django, implementing user authentication, music playback, playlist management, and CRUD-based content management.
- Built 10+ responsive pages with reusable Django templates, enabling users to browse tracks, create and manage playlists, and stream music through an integrated web player.
- Implemented secure authentication and user-specific data handling, connecting frontend interfaces with Django views, models, and database operations.

## Achievements

- **Winner – AI/ML Track, KU Hackfest 2025** [Certificate]
- **Student Ambassador – GeeksForGeeks Campus Mantri 2026** [Certificate]
- **Student Partner – Internshala, ISP-2026** [Certificate]
- **2nd Runner-Up – Hult Prize at KhEC 2023** [Certificate]

## Leadership and Extracurricular

- Strengthened cross-functional delivery on Agile/Scrum teams, spanning UI/UX, backend integration, and project coordination responsibilities.
- Tackled real-world security vulnerabilities in the Pentester Nepal PMO-Sponsored Challenge, applying practical penetration testing methodologies to solve complex cybersecurity problems.
- Competed in the NCA Hack The Box CTF Challenge, successfully capturing both the **user** and **root** flags by troubleshooting systems and exploiting infrastructure vulnerabilities under time constraints.
- Sharpened collaborative problem-solving under real-world constraints, by actively participating in hackathons, ideathons, and competitive technical challenges.
- Served as **Vice President**, IT Circle – Khwopa Engineering College, leading committee initiatives and coordinating technical event planning.
- Produced promotional materials and visual assets for college technical events and club activities, as **Graphic Designer** for IT Circle, Khwopa Engineering College.